



Air University, Islamabad

Project Proposal

for

Bionic Vision System

Version 1.0

By

Yousbah Zahid 231231

Yumna Hussain 231149

Seerat Aslam 231167

Supervisor

Sir Adil Sheraz

Co-Supervisor

Mam Qurat-ul-ain

Bachelor of Science in Artificial Intelligence (2023-2027)

Table of Contents

Abstract.....	3
1. Introduction.....	4
2. Problem Statement	4
3. Problem Solution/Objectives Of The Proposed System	4
3.1 Objectives	5
4. Related System Analysis/Literature Review.....	5
5. Vision Statement	5
6. Scope	6
7. Modules	6
Module-1: Data Acquisition And Preprocessing	6
Module 2: Hazard Detection.....	7
Module 3: Dynamic Obstacle Detection & Tracking	7
Module 4: Proximity Estimation And Approach Detection	9
Module 5: Risk Assessment & Situation Understanding.....	10
Module 6: Decision & Instruction Generation Module	11
Module 7: Warning Sign Understanding	12
Module 8: User Interaction & Voice Execution Module.....	13
Module 9: Manual Interaction	14
Module 10: Macro Navigation.....	15
Module 11: System Integration & Mobile Processing Architecture	15
8. System Limitations/Constraints.....	17
9. Tools And Technologies.....	18
10. Project Stakeholders And Roles	19
11. Module Based Work Division	19
12. Wbs And Gantt Chart.....	20
13. Mockups	24
14. References.....	25
15. Plagiarism Report.....	25

Project Category: (Select all the major domains of proposed project)

- | | |
|--|---|
| ☐ A -Desktop Application/Information System | ☐ B -Web Application/Web Application based Information System |
| ☐ C -Problem Solving and Artificial Intelligence | ☐ D -Simulation and Modeling |
| ☐ F -Smartphone Game | ☒ E -Smartphone Application |
| ☐ Other (specify category) | ☒ H -Image Processing |

Abstract

Millions of visually impaired individuals face life-threatening challenges while navigating the challenging spaces of urban settings and street environments that lack intelligent, real-time assistive technology. Current technologies such as the white cane, Envision Glasses, Seeing AI, Google Lookout and guide dogs, are costly, provide limited situational awareness and existing smartphone apps provide only turn-by-turn directions at a macro-level, do not detect obstacles at the ground-level such as potholes, vehicles and others. The Bionic Vision System aims to develop a low-cost, wearable smart glasses-based assistive solution that is integrated to a mobile application. The System detects static and dynamic hazards, estimates distance and provides audio guidance at micro-level while following the direction of macro-level map routes. The System also includes context-aware Optical Character Recognition (OCR) for extraction of safety-related information from the environment.

1. Introduction

“Bionic Vision System” is an innovative wearable assistive technology to enhance independent mobility, safety and spatial awareness of people with blindness/low vision. Existing mobility aids lack real-time awareness for navigating dynamic, cluttered, crowded, unfamiliar indoor/outdoor settings. Our implementation is to bridge the gap between conventional tools and modern AI by developing smart edge-to-mobile vision system.

The main system comprises a set of lightweight smart glasses integrated with an on-board camera sensor and a high-performance smart phone application. The Bionic Vision System utilizes deep learning algorithms, computer vision such as real-time object detection, depth estimation and Optical Character Recognition (OCR) to continuously analyze the environment around the user and convert it into useful visual information (auditory signal). This document serves as the initial design description of the hardware aspect of the project explaining functional architecture, various system modules and required specifications to be met for implementation.

2. Problem Statement

Visually impaired individuals struggle daily while walking independently in dynamic environments. While traditional tools like the white cane Envision Glasses, MS Seeing AI, OrCam, weWalk, Google Lookout are helpful, they have major limitations. A white cane and weWalk can only detect obstacles on the ground within a very short physical distance. OrCam and Envision Glasses are too much expensive like 4000\$ which cannot be afforded by normal person, Seeing AI and lookout do not use any specific trained CV models and required mobile must be in hand mostly completely fails to warn the user about hanging objects (like tree branches, wires), fast-moving hazards (like cars or bikes), or read important warning signs on the road. This is unsafe and causes stress or visually impaired person to walk in unfamiliar or crowded environment. Furthermore, currently available smart navigation devices are more likely to fail in real-world scenarios. Many of these systems rely on cloud computing, due to which the experience high latency and if the internet connection stops complete system will stop working. On the other hand, systems that try to process AI locally with the help of microcontroller reduce the resolution too much hence decreasing the accuracy of the system. Therefore, there is a critical need for a fast, cost effective, and easy to use solution. The visually impaired requires a system that can detect object without any latency and detect the object from far, process visual data and generate an audio feedback.

3. Problem Solution/Objectives of the Proposed System

Most of the time visually impaired people require some caregiver for their assistance. They do not require a caregiver or a stick for guidance but a smart guide. We are building a smart pair of glasses that connects to the user's phone and our guide app will assist them in the best possible way.

Smart Wearable Glasses:

A camera will be attached to the glasses which continuously records the environment in real time and a button will choose the environment mode (e.g. indoor, outdoor). This visual data serves as the primary input for the system to understand the user's surroundings.

The Virtual Guide:

This system bridges gap between **Macro-Navigation** and **Micro-Navigation**. Macro navigation gives a big picture like the GPS guiding through the streets whereas micro navigation focuses on tiny details like potholes, or manholes.

Advanced Perception:

It spots static and dynamic hazards, such as poles, potholes, and moving cars. This system employs YOLO based object detection for such hazards. Instead of using expensive hardware, the system uses a single camera to estimate depth, calculating exactly how far away a danger is to judge the risk of a collision. Finally, it uses context-aware OCR to read only the most important signs, like "Warning: Construction" or "Bus Stop," providing clear audio instructions to guide the user safely.

Smart Audio Feedback:

The system uses priority based audio logic instead of stream of data. This prevents from information

overload and alerts user with necessary navigational cues and immediate threats.

3.1 Objectives

Comfortable Design:

A smart pair of glasses that are easy to carry, having camera attached to it which captures live video.

Real Time Hazard Detection:

To implement and optimize deep learning models like YOLOv8 to detect objects either ground level (e.g. potholes) or hanging objects (e.g. wires) with a minimum processing speed.

Depth Mapping:

The system uses the technique of depth estimation mapping the distance between the user and the obstacle (static or dynamic). It provides user spatial awareness determining if an object is an immediate threat or a safe distance away without the need for expensive sensors.

OCR Integration:

This system understands the environmental context along with textual signs. It interprets the overall scene, differentiating between a sidewalk, a crosswalk, or a store entrance. Optical Character Recognition (OCR) is used to read important text from signs, like "Bus Stop," "Warning: Construction," or store names.

Low Latency Mobile Integration:

To create a mobile app that acts as the system's brain, quickly processing everything the camera sees without any lag or delays. It serves as the central processing unit, ensuring seamless data transfer between the glasses and the AI models via a high-speed wireless connection.

Voice Guidance:

To design a text to speech (TTS) interface which provides commands based on environmental analysis. It is a simple voice system that gives clear, step-by-step instructions like "Stop, car approaching" or "Turn left," making navigation easy and stress free.

4. Related System Analysis/Literature Review

Application Name	Weakness	Proposed Project Solution
Drishitikon: An advanced navigational aid system for visually impaired people	Use Raspberry Pi and often external power banks. This makes system expensive.	All navigation is done through simple RGB camera which decrease high cost of the glasses.
Drishiti: Visual Navigation Assistant for Visually Impaired	It's navigation system does not provide macro navigation to reach any specific destination.	Provide macro navigation using map api and connect it with micro navigation system.
A Convolutional Neural Network based Live Object Recognition System as Blind Aid	Don't provide depth estimation due to which distance of hurdle from user can't be determined.	Have depth estimation module to provide timely warning about hurdle.

Table 1 - Related System Analysis with Proposed Project Solution

5. Vision Statement

For visually impaired individuals **who** wants to walk safely in any indoor or outdoor environment without taking assistance from any caregiver, **the** Bionic Vision System **is** an affordable solution of a pair of smart glasses and a mobile app **that** bridges the gap between macro level GPS routing, and micro level hazard detection by providing real time audio guidance for static, dynamic, and ground level hazards. **Unlike** expensive tools like Envision AI or OrCam which lacks navigation, dynamic obstacle detection or white canes that miss objects hanging like wires, **our product** incorporates depth estimation and context aware OCR into a single, low latency pipeline to guarantee safe,

independent mobility without the need for constant caregiver.

6. Scope

The “Bionic Vision System” is a smart assistive navigation platform consists of two major components: a glasses equipped with camera ESP32 and an Android mobile application. The scope of this project involves the complete development and integration of eleven functional modules that together provide real-time AI-based navigation assistance for visually impaired users. The system captures video with the help of camera and pass it to multi stage AI pipeline that runs on mobile app. The data acquisition module is used to capture video, extract frames, normalize and optimize it to run in real-time. The hazard detection module will detect at least six types of static ground-level barriers such as potholes, open drains, construction debris, electric wires, animals, and sidewalk edges, using a deep learning model based on YOLOv8. The dynamic obstacle tracking module will identify and follow moving objects like vehicles and pedestrians through video frames to estimate their velocity and direction in order to predict the risk of collision. To estimate relative distances of objects detected, the depth estimation module will use a monocular depth model to compute the relative distance of the objects detected by the system and classify them into close, medium, and far proximity zones. The risk assessment module will integrate outputs of the detection, tracking and depth modules to calculate a risk score of each hazard and rank alerts based on this risk score. The decision and instruction generation module will be used to translate the risk assessment into brief, actionable text commands that will then be converted to voice output via a text-to-speech engine. The OCR and scene understanding module will be able to read safety relevant text in the environment using Easy OCR and only provide it to the user when it is contextually appropriate. The user feedback module will take care of voice alert delivery, priority based announcement ordering and optional haptic feedback. It will also be compatible with the current map APIs to add micro-level navigation with macro-level route guidance. It is restricted to outdoor city settings and will be tested using real-life controlled experiments. Hardware production is not part of the project; an already existing wearable camera in the market will be modified to fit within the project. Cloud based processing will not be available, all AI inference will occur on the mobile device itself to guarantee low latency and offline functionality.

7. Modules

MODULE-1: DATA ACQUISITION AND PREPROCESSING

FE-1: Live Stream Receive

ESP32-S3 connected to laptop, code run on Arduino, Wifi connects, IP address will be generated, paste it on browser and start streaming (this is only done once). Next time, glasses are put on, streaming automatically starts (App on, connects to Wifi, streaming starts). This is a continuous flow of frames.

FE-2: Frame Rate Control

User’s surrounding does not change drastically, therefore, reducing frames will reduce the latency and increase processing. ESP32-S3 (with OV2640 camera) can deliver upto 15 frames per second. If we try to process all frames, latency will occur. Consider there is a manhole in frame 1, but person already passed it because frame 2, 3, 4.... are still processing. Frames can be reduced by discarding every second or third frame. This ensures real time responsiveness and low latency scene understanding.

FE-4: Resize to 640x640 (Letterboxing)

OV2640 delivers frame typically 640x480 pixels that is a rectangular shape. YOLO model requires 640x640 input (square shaped). If the frame is not resized and passed to model, it may crash or give incorrect results. We will use letter boxing technique to resize the image, extra pixels will be shaded black or white, this will not stretch the original image and proportions of environmental hazards will not be destroyed. Consider a manhole is in round shape, normally resizing will distort its shape and will appear oval.

FE-5: Pixel Normalization

YOLO understands data scaled between 0.0 and 1.0. Images are scaled from 0 to 255 to represent colors. These pixel values are normalized by dividing each pixel value by 255.0. This way data will be scaled between 0.0 and 1.0.

MODULE 2: HAZARD DETECTION

FE-1: YOLOv8 TFLite inference

YOLO is a method that helps in object detection. Raw images are trained on YOLOv8 model and a .pt file is created which is converted to .tflite file. This is done because .pt file is heavy and slow. When converted to .tflite file, that heavy file is compressed and fast. TFLite provides the speed and efficiency while YOLO acts as logic that scans image. According to this, a processed frame is received from module 1 to model where it detects the object and tells its class, confidence score and bounding box.

FE-2: Confidence Filter

Confidence score of objects vary. In an image there is a shadow, and YOLO model is not fully sure about it and detects as person so its score will be low. So, only those objects will be focused having confidence score 0.5 (model is 50% sure) or higher, everything below 0.5 will be discarded. This 50% surety is kept as threshold. If it is low, it could lead to constant false alarms. If it is high, it may skip real obstacles.

FE-3: Mode based class filtering

Even after confidence filtering, not all detected objects are hazards. For example, if a sofa is detected in a room, it is not an obstacle. Similarly, if a model detects car and the user is sitting in his room, this is a wrong detection as cars are not present in rooms. So, we use environment mode technique, mode is selected like indoor, outdoor and only relevant classes will be focused on. This is used to filter objects by classes. Consider, user selected Indoor room mode (by pressing button once), stairs, walls, and doors are treated obstacles while furniture like chair, and sofa are ignored as they are not harmful at home. Similarly, in corridor mode, list will expand and furniture will also be obstacle but potholes, manholes will be ignored as they are not present in hall pathways or corridors. When outdoor mode is selected, all things become hazards and list will expand to potholes, wires, speed bumps and all street hazards.

FE-4: Top 1 priority selection

After all filtering, there still is chance that there are multiple objects in list. If all are considered and alerts are generated for all (e.g. “car is on right, dog is on left, bike ahead”), this will panic the blind person and he/she will not get what to do. Therefore, bounding box area is calculated for each object, the object with largest area will be prioritized and alert will be generated for that first.

MODULE 3: DYNAMIC OBSTACLE DETECTION & TRACKING

3.1 Phase 1: Identifying the Movers:

This phase act as a “eye” for our navigating system. It scans the live video coming from the camera and finds the moving objects that could be dangerous to the visually impaired people.

- **Primary Detection Engine:**

We use YOLOv8 model which is a pre-trained model on COCO dataset used to detect moving objects in real-time by looking at the camera frame.

- **Smart Class Filtering:**

The standard YOLO model can detect over 80 different objects, we don't need all of them so we wrote a custom python code to filter out useless or safe classes like the flying bird or airplane.

FE-1: Vehicle Detection:

The system specifically looks at high speed vehicles like car, bus or motorbike that are near the user's walking path.

FE-2: Pedestrian Detection:

The system also detect the people walking nearby to prevent the use from colliding into the people in crowded streets

3.2 Phase 2: The Tracking Engine

Once the moving objects are identified this phase acts as a memory. It keep tracks of how fast the object is moving and where the object is moving.

FE-3: Object Tracking:

Standard YOLO model is memoryless means it treat each frame independently means it does not have memory of previous frames, to overcome this problem we will use **DeepSort** which is popular CV algorithm for multi-object tracking it assigns Unique id to each object and it uses Kalman Filter that predicts the future position of the currently tracked objects based on their previous velocity and motion.

FE-4: Speed Approximation:

Since we are using only camera no extra sensor or radar for speed calculation so to approximate speed we will use frame's time and distance known as temporal frame displacement. By the concept of **Bounding Box Analysis** DeepSORT uses Kalman Filter and analayz the YOLO boxes to detect the speed and direction of the object If the box moves Left/Right, the object is crossing the path and If the box gets rapidly larger (expands in width/height), the object is moving directly toward the user at high speed.

3.3 Phase 3: Risk Prediction & Alert Trigger

This is the final decision maker phase of the module. In this phase the tracking and speed of the object is combined to check whether the object is really dangerous or not.

FE-5: Direction Estimation:

The system uses temporal frame comparison to identify the exact direction of the moving object. It simply identifies whether the object is moving left or right or moving away from the person, which indicates a safe situation.

FE-6: Approaching Object Identification & Ego-Motion Compensation:

A major challenge in mobile vision is distinguishing between a stationary object (like a parked car) that appears to grow larger simply because the user is walking towards it, and an actively moving hazard heading toward the user. To solve this, the system does not just look at the size increase; it calculates the Rate of Expansion (RoE) of the bounding box over consecutive frames. The system applies a mathematical threshold to calculate the relative expansion speed:

$$\text{RoE} = \frac{\Delta \text{Area}}{\Delta \text{Time}}$$

Equation 1 - Rate of Expansion

Since a visually impaired person walks at a slow, predictable speed, a stationary object's bounding box will expand gradually. If the calculated RoE falls within this normal walking threshold, the system tags the object as static. However, if the bounding box expands rapidly and violently—exceeding the walking threshold—it mathematically proves that the object has its own forward velocity and is actively moving towards the user. In such cases, the system instantly classifies the object as a dynamic collision hazard.

- **High-Risk Alert Generation:**

After identifying the threat the module immediately a High Risk status flag along with object class and ID to the module 5(Risk Assessment).

MODULE 4: PROXIMITY ESTIMATION AND APPROACH DETECTION

FE-1: Receiving Bounding Box Data

Module 4 receives its input from Module 2 and Module 3. Module 2 provides bounding box coordinates and class names of detected static hazards. Module 3 provides bounding box coordinates, unique tracking IDs, approach speed flags, and direction information of detected dynamic objects. Module 4 does not perform any detection of its own, it only performs proximity calculations on the data received from these two modules.

FE-2: Area Ratio Calculation

Using the bounding box coordinates received from Module 2 and Module 3, the area ratio is calculated. The formula is $\text{box_area} = (x_2 - x_1) \times (y_2 - y_1)$, $\text{frame_area} = \text{frame_width} \times \text{frame_height}$, and $\text{area_ratio} = \text{box_area} / \text{frame_area}$. This ratio falls between 0 and 1. A small ratio indicates the object appears small in the frame and is far away while a large ratio indicates the object is close. This value serves as the primary proximity indicator for zone classification.

FE-3: X and Y Ratio for Reliability Check

Area ratio alone becomes unreliable when the camera angle changes or when an object is only partially visible in the frame. To handle this, $x_ratio = (x_2 - x_1) / \text{frame_width}$ and $y_ratio = (y_2 - y_1) / \text{frame_height}$ are calculated separately. If y_ratio is less than 0.15 and x_ratio is greater than 0.35 simultaneously, the glasses are tilted downward and area ratio cannot be trusted. In this case x_ratio is squared to produce an $x_based_proximity$ value which is used instead. If the object center is within 80 pixels of the left or right frame edge the object is partially visible from the side. If center_y is less than 80 pixels the object is entering the frame from above. These situation flags are included in the output packet so that Module 5 accounts for visibility conditions during risk scoring.

FE-4: Class Specific Thresholds and Zone Classification

Different object classes have different real world sizes so a single threshold for all classes would produce incorrect proximity readings. A bus far away can have the same area ratio as a pothole that is very close. To handle this each class is assigned its own near and medium threshold. Person uses near 0.10 and medium 0.05. Car uses near 0.15 and medium 0.07. Motorcycle uses near 0.12 and medium 0.06. Bus and truck use near 0.20 and medium 0.10. Pothole and manhole use near 0.03 and medium 0.01. The proximity value is compared against these thresholds to assign a zone of NEAR, MEDIUM, or FAR. Objects in the FAR zone have a proximity value too small to pose any immediate threat and their risk score in Module 5 will automatically remain low, producing no alert.

FE-5: Direction Calculation for Static Hazards

Module 3 already provides direction information for dynamic objects through its tracking pipeline. For static hazards detected by Module 2, direction is calculated in this module. The frame is divided into three equal vertical zones. X_{center} is calculated as $x_1 + (x_2 - x_1) / 2$. If X_{center} falls between 0 and 213 the object is on the LEFT. If it falls between 213 and 426 the object is ahead in the CENTER. If it falls between 426 and 640 the object is on the RIGHT. This direction value is included in the output packet for static hazards and is used by Module 5 and Module 6 to generate appropriate navigational instructions.

FE-6: Speed Flag Integration for Zone Adjustment

Module 3 provides approach speed flags for every tracked dynamic object; these are STATIONARY, APPROACHING, FAST APPROACHING, and MOVING AWAY. Module 4 combines these speed flags with the calculated proximity zone to produce a more accurate picture of the threat level before passing data to Module 5. If

an object is in the MEDIUM proximity zone but Module 3 has flagged it as FAST APPROACHING, this combination is marked with high urgency in the output packet so that Module 5 assigns it a higher risk score than a stationary object in the same zone. A stationary object in MEDIUM zone receives no urgency flag. This adjustment ensures that a fast moving motorcycle at medium distance is treated more seriously than a parked car at the same distance.

FE-7: Object Level Classification

The vertical position of the bounding box combined with the object class name determines whether the object is a ground level hazard, an overhead hazard, or an eye level object. Classes such as pothole, manhole, undercover manhole, open drain, speed bump, and debris are always classified as GROUND. Classes such as wire, branch, and rope are always classified as OVERHEAD. For any other class if y1 divided by frame height is less than 0.15 the object is entering from above and is classified as OVERHEAD. All remaining objects are classified as EYE_LEVEL. This classification is forwarded to Module 5 because ground and overhead hazards carry different danger weights in the risk scoring formula.

FE-8: Final output packet to module 5

After all the calculations and cool down check passes, module 4 sends this data packet containing the object's class name, distance zone, direction, and speed flags to next module 5.

MODULE 5: RISK ASSESSMENT & SITUATION UNDERSTANDING

This module act as a brain for the navigating system. It gets the data from Hazard detection, Tracking and Depth estimation module to calculate how dangerous the situation is and make logical decision to keep the user safe.

5.1 Waking Path Analysis:

The camera view is divided into three main zone ie; left, right and center. If the object is spotted at center it will be treated at much higher risk then the objects at left and right.

5.2 Risk Scoring Formula:

The system will use a risk scoring formula to give "Risk Score" to every detected object based on its type and distance from the user.

$$\text{Risk Score} = (W_{\text{class}} \times \frac{1}{D}) + M_{\text{penalty}}$$

Equation 2 - Risk Score

Every object will have its pre-defined danger value which is its weight W_{Class} for example an open drain might have weight 0.9 and a parked car may have weight 0.4.

D (Distance) is the estimated distance from the module 4. As the distance is in denominator when the distance is less the risk score will automatically shoots up.

M_{penalty} (Motion penalty) is the user tracking from module 3, if the object is moving rapidly toward user extra penalty score will be added and in the case of static object this penalty will remain 0.

5.3 Hazards Prioritizing Ranking:

In complex urban areas the camera will likely detect multiple obstacles simultaneously. Generating audio for every single object will cause confusion and irritation for visually impaired people to solve this problem the system implements a **prioritization queue**.

By using the calculated Risk Score from previous step the system sorts all the detected hazard in descending order. The system will isolates the object with highest risk score warn the user about it first and once this threat is bypassed the system evaluate the next highest threat in the queue.

5.4 Confidence-Aware Filtering (False Alarm Reduction):

The models give output prediction with some probability percentage known as “Confidence Score”. In outdoor environment the factors like shadow, blurry frames and poor lightning can cause the model to falsely detect hazard with a low confidence score and generating a sound. To prevent this Alarm Fatigue the system sets a pre-defined confidence threshold any hazard below this threshold will be discarded as false positive. This mechanism only generate a alert for high probability hazards, thus keeping the user focused, calm and trustful of the navigation system.

Output of the Module: After processing all the data through risk score and confidence filter the module finalize the actionable threat. Now this treat is forwarded to module 6 (Decision Making & Audio Feedback) to generate an immediate navigational voice instruction for the user.

MODULE 6: DECISION & INSTRUCTION GENERATION MODULE

This module serves as a communication system for our navigation system. Its primary purpose is to get data from module 5 and generate a safe real-time instruction for the visually impaired user.

6.1 Path Guidance:

Instead of announcing the name of the detected object, the system utilizes the decision tree to determine the safest action based on threat’s location. It generates safe path suggestion by instructing user to move left, move right or stop. How decision Tree logic works:

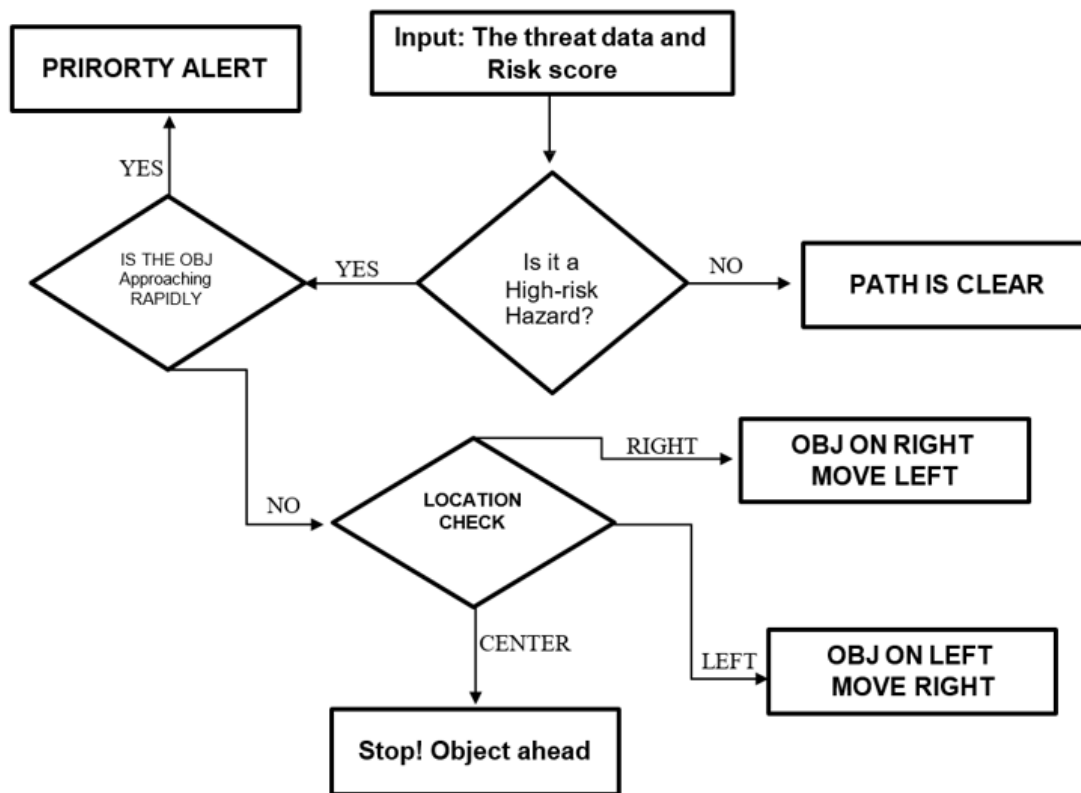


Figure 1 - Decision Tree Logic

6.2 Conflict Resolution:

As discussed in Module 5 the camera will likely detect multiple obstacles simultaneously, to prevent confusion and irritation for the user this feature provides conflict resolution between multiple hazards. It compares the prioritize “RISK SCORE” from module 5, the system suppress the low score threats and isolates the high score threat. This

ensures the user receives only one clear, uncompromised safety instruction at any given moment. In a rare moment in which the obstacles have same highest risk score, the system employs a Tie-Breaker protocol. It defaults to 'Zone Priority', prioritizing the hazard in the 'Center' path over Left/Right threats. If both threats are at the primary path, the system defaults to a 'Fail-Safe' mode, instantly issuing an overriding "Stop! Multiple hazards" command to ensure maximum user safety.

6.3 Instruction Stability & Cooldown Protocols:

One big problem with real-time is something called "state flickering". This happens when the camera shakes or an object moves slightly and the system keeps switching between different commands really fast. To stop this from happening the module uses something called Instruction Stability Control. This feature puts a kind of time delay on a command so the system does not keep changing commands.

The system also has a problem with making much noise. If it keeps beeping all the time it can be really annoying and tiring for user. So the system uses a Time-Based Cooldown Mechanism. When it sends out a warning it waits for a while before sending out the same warning again. This stops the system from making much noise.. If the tracking module sees that something is getting closer or moving faster really quickly it will send out a special emergency alert right away even if the cooldown time is not up yet. The system uses Instruction Stability Control to make sure this happens smoothly. The Time-Based Cooldown Mechanism is also important, for preventing the system from making much noise and it works well with Instruction Stability Control to make the system more useful.

6.4 Real-Time Updates & Low-Latency Speech Output:

As the user move the system updates its decisions in time to match any sudden changes in the surroundings. It can even change its instructions if needed.

To ensure maximum safety, the system integrates a low-latency Text-to-Speech (TTS) engine. When the system decides on a navigation instruction it quickly turns the text into speech. This way you get voice guidance without any delay, which is really important, for staying safe when there are fast-moving dangers around.

MODULE 7: WARNING SIGN UNDERSTANDING

This module deal with understanding of text on warning signs and classifying into predefined hazard category to generate necessary real-time voice based instruction.

FE-1: Warning Sign Detection

Detect warning sign boards using object detection module.

FE-2: Optical Character Recognition (OCR) Triggering

The OCR is only triggered when a valid warning sign board is detected. Sub-processes:

Ratio of sign board area to frame area is calculated. Image sharpness using laplacian method is calculated. Persistence of sign board across various frames is determined. OCR is triggered when all readability conditions are met.

FE-3: Multilingual Text Processing

Both English and Urdu text are processed for warning sign understanding.

FE-4: Warning Text Identification

Warning text is identified using semantic similarity with reference embeddings. Sub-processes:

Reference embeddings dataset is created of warning text samples. Extracted Text is converted to embeddings. Cosine similarity with reference embeddings are calculated. Text is classified as warning text if similarity scores exceed certain threshold.

FE-5: Classification of Warning Category

Warning text is classified into type of warning like road closed, wet road, road closed, restricted entry etc. Category is determined on the basis of highest similarity scores between all respective classes.

MODULE 8: USER INTERACTION & VOICE EXECUTION MODULE

This module act as communication layer between user and bionic vision system. It is responsible for giving audio instructions to user and also receiving audio commands from the user.

8.1 Real-Time Audio Instructions

This part is responsible for providing real-time speech output.

FE-1: Text to Speech

This feature use Android text-to-speech to convert textual instructions given by system into audio output.

FE-2: Multilingual Output

This feature can provide output in language according to user preference.

8.2 User's Audio Commands

This feature allows user to control system with their voice command.

FE-3: Speech Recognition

This uses speech-to-text to convert user's command into textual instruction for further processing. Supported commands include:

- Repeating last instruction.
- Volume change.
- Mute / Unmute
- Language Change

FE-4: Multilingual Commands

This feature is responsible for understanding user command in both Urdu and English based upon system settings.

FE-5: Repeat Instruction

This feature operate on command 'Repeat' and it does repeat the last instruction given by system.

FE-6: Language Switching

It switches languages based on fixed commands "Bionic Vision English" and "Bionic Vision Urdu".

FE-7: Volume Change

This changes volume up and down based upon fixed pre-defined "Volume Up" and "Volume Down" commands.

FE-8: Mute / Unmute Alerts

This command blocks non-critical instructions temporarily. Beneficial for navigating in very crowded places.

FE-9: Navigational Commands Interpretation

It can interpret navigation related voice commands and then forward them to macro-navigation module.

Example:

Asking about nearby coffee shops.

Generating route to school

8.3 Priority Hierarchy of Instructions

Voice execution module receives instructions from multiple sources. To ensure that most important instruction is delivered first priority based hierarchy is built.

Highest: Micro-Level Navigation

Instruction generated upon the basis of risk assessment of hazards. e.g., ‘stop’ , ‘move left’.

Medium: Macro-Level Navigation

Instruction given by map API toward the destination. e.g., ‘take right turn onto street 9’.

Lowest: User Interaction Command

These are on the basis of user command. e.g., REPEAT

8.4 Parallel Execution

To ensure smooth working and efficiency this is important.

FE-10: Concurrent Execution Management

This feature manages multiple processes parallelly.

User voice command listening.

Audio output processing.

Navigation instruction from map API

8.5 Instruction Interruption

Sudden hazard require immediate responses in real-time. This feature is for interruption of low-priority instruction.

FE-11: Instruction Interruption Mechanism

In case of high priority instruction is generated ongoing speech is interrupted immediately to deliver the high priority one.

FE-12: Audio Playback Control Layer

This control speech execution by managing its start, end and is responsible for preventing speech overlap.

MODULE 9: MANUAL INTERACTION

In case of loud surrounding noises voice commands are not possible. This feature provide way to provide non-verbal communication between user and system. It uses button combinations and motion-based gesture to provide commands.

FE-1: Hardware Based Volume Control

Uses simple ‘voice up’ and ‘voice down’ buttons to increase or decrease the voice of instructions provided.

FE-2: Hardware-Based Mute / Unmute

It is activated when both volume buttons are pressed simultaneously. In active audio → Enters mute mode.

In inactive audio → Enters unmute mode.

FE-3: Hardware-Based Repeat

When any of the two volume buttons are pressed for 3 seconds “REPEAT” command is triggered and most recent instruction is repeated.

MODULE 10: MACRO NAVIGATION

This module deal with macro navigation provide route guidance from origin to destination. It focuses on destination tracking and provides all map-based features to user according to user requirements.

10.1Nearby Destination Discovery

This allow users to search nearby places and select destination from them.

FE-1: Location Detection

This feature uses GPS to determine the current location of user.

FE-2: Point of Interest

It searches for nearby places according to user request like grocery stores, hospital and banks etc. Then output the desired places along with their distances to user in audio.

10.2Route Generation and Guidance

This generate route for user according to specified destination.

FE-3: Pedestrian Route Generation

This generate fastest and pedestrian friendly map for user to selected destination.

FE-4: Navigation Instructions

It provides step by step instruction to user at each route checkpoint. It extract each instruction from full route. It determine checkpoint by tracking GPS of user.

FE-5: Route Tracking

Route is continuously tracked starting from origin. If deviation from determined route is detected a new route is calculated.

10.3: Integration with Micro-Navigation System

It is connected with micro-navigation to provide a safe hazard free guidance.

FE-6: Priority Based Navigation

Macro-navigation is of less priority than micro-navigation and its instructions are always delayed in case of micro level instruction.

MODULE 11: SYSTEM INTEGRATION & MOBILE PROCESSING ARCHITECTURE

11.1Hardware:

The system uses ESP32-S3 high speed sensor node. It captures the video via the OV2640 camera, optimizes the frame rate (FPS) then transfers the low latency video stream to mobile application with the help of WIFI connection.

11.2Mobile processing & TF-Delegates:

The mobile application serves as a “Brain” for the system. To process video frames from camera without causing battery to drain and phone to heat, the application uses TensorFlow Lite (TF-Lite) and Hardware Delegates. The application dynamically distributes the model between mobile

device's CPU, NPU and GPU.

The tasks like hazard detection (Module 2) and dynamic obstacle detection (Module 3) are delegated to NPU for extra fast processing. Depth Estimation utilize the GPU, while the intermittent tasks like OCR are processed by the CPU. This parallel hardware guarantees High FPS execution of all models concurrently and ensures long range detection plus immediate audio feedback.

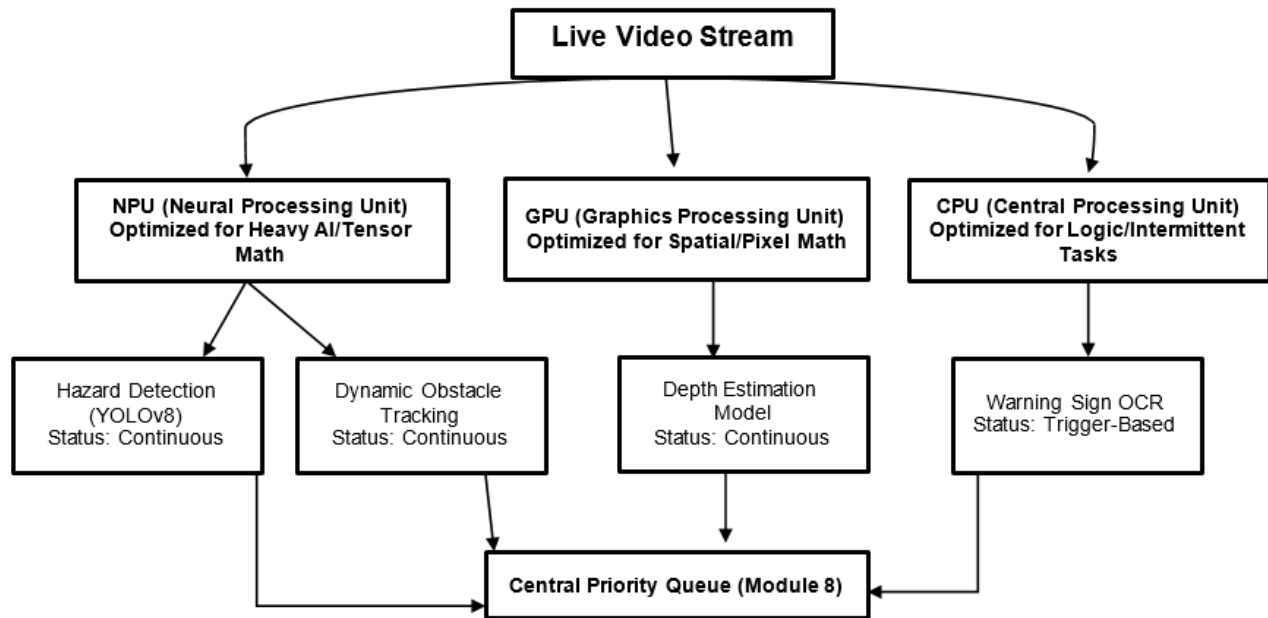


Figure 2 - Heterogeneous Distribution

11.3 Continuous Background Multitasking:

To ensure safe user experience the app will run continuously in the background even when the phone is locked unless the app is removed from background.

11.4 Instant Audio Alert & Screen Free interface:

Once the hazard is detected the app uses phone built in TTS to speak the warning aloud. Since all the AI processing is done using user's high-end device no internet is used for processing the voice warning will be delivered without any delay.

Moreover the system is designed for visually impaired so there is no need of visually complex screen layout, the app features a SCREEN FREE design where the user interact entirely through voice commands, phone vibrations and beeps.

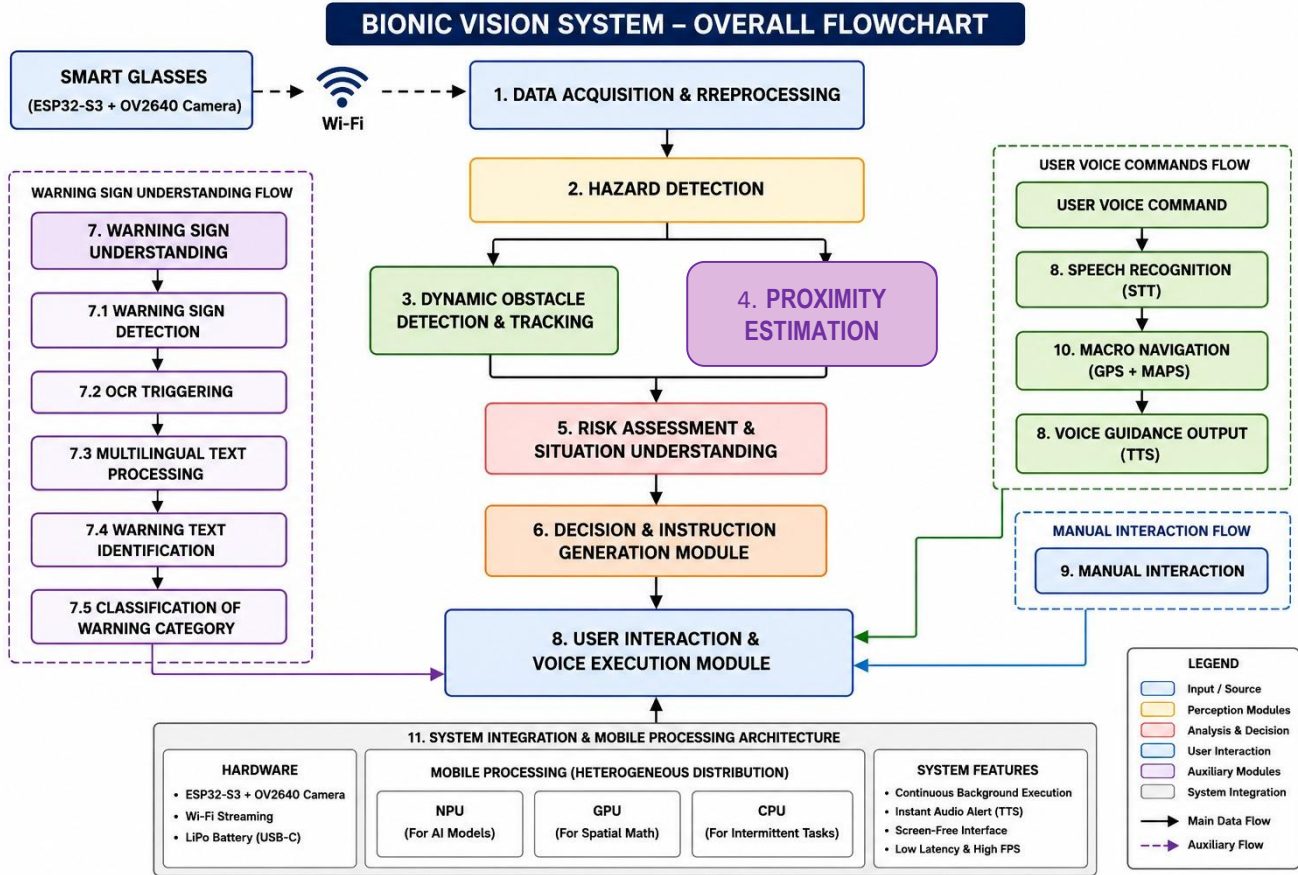


Figure 3 - WORK FLOWCHART

8. System Limitations/Constraints

Although the suggested navigation system is highly-advanced in its assistance, it has some practical limits and constraints:

- **Lighting Dependency:** Since the system uses a standard camera sensor to observe the world, the system requires good lighting in order to work effectively. It will either find it very difficult to see in environments unless it has an external flashlight.
- **Battery Usage:** It is an intensive burden to run several AI deep learning models at a given time. Although workload is shared by defining hardware delegates, the system will consume the smartphone battery much more than daily usage. Before such lengthy walks, the user should make sure that their phone is charged adequately.
- **Hardware Requirement:** It is the main brain of the system as the mobile application. Thus, it will not run well with very old or low end smartphones. It stipulates a modern, mid-to-high-end smartphone with decent processing power to execute the AI models without freezing.
- **Weather and Physical Obstructions:** Accuracy of the system is dependent on a clear sight. The camera lens can be blurred by extreme weather conditions such as heavy rains or thick fogs. On the same note, in the event that the user accidentally covers the camera with his/her hand or clothing, the system will temporarily cease to give instructions.

- **Wireless Range Limitation:** The smart glasses and the mobile phone will also need to be connected locally with a local wireless connection (Wi-Fi/Bluetooth). The user should have the smartphone physically attached to him or her (in his or her pocket or bag). And when the phone is left in a different room the glasses will disconnect and cease to work.

9. Tools and Technologies

Table 2 - Tools & Technologies for Proposed Project

Category	Technologies	Version / Specification	Rationale
Machine Learning	YOLOv8	Pre-trained object detection model on COCO dataset	For hazard detection
	DeepSORT	Pre-trained CV Algorithm	Use to calculate future position of object detected on the basis of it's current position and velocity with the help of Kalman's filter
	TensorFlow Lite (TFLite)	ML Framework mobile based	Use to run machine learning models on mobile phone.
Hardware & Sensors	XIAO ESP32-S3 Sense / ESP32-CAM	Xtensa Dual-Core 32-bit LX7 CPU, 240 MHz	Microcontroller board capturing live video through the OV2640 sensor.
	OV2640	Camera Module with Ribbon Cable	Captures 640p/480p environmental frames and streams them over Wi-Fi/BLE.
	LiPo Battery	USB-C Rechargeable	Use for providing power for the smart glasses.
Text and Speech Processing	Android Text-to-Speech (TTS)	Native Device Library	Convert textual instructions into audio instructions

10. Project Stakeholders and Roles

Table 3 - Project Stakeholders and Roles

Role	Name / Entity	Responsibility
Project Sponsor	Air University, Islamabad	Institutional support, lab resources, and project evaluation framework
Primary Users	Visually Impaired Individuals	End users of the system; source of requirements through interviews and user testing
Team Member 1	Seerat Aslam (231167)	AI Model Development: Hazard detection, depth estimation, and risk assessment modules
Team Member 2	Yousbah Zahid (231231)	System Integration & Tracking: Dynamic obstacle tracking, decision engine, and mobile application
Team Member 3	Yumna Hussain (231149)	Data & Output Layer: Data preprocessing, OCR module, voice feedback, and dataset preparation
Project Supervisor	Mr. Adil Sheraz	Technical guidance, milestone evaluation, and proposal approval
FYP Committee	Air University FYP Committee	Final evaluation and grading of the project deliverables

11. Module Based Work Division

Table 4 - Team Members Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Module / Feature
Seerat Aslam	231167	Module 1 (Data Acquisition and Preprocessing), Module 2 (Hazard Detection) and Module 4 (Proximity Estimation)
Yousbah Zahid	231231	Module 3 (Dynamic Obstacle Detection & Tracking), Module 5 (Risk assessment & Situation understanding), Module 6 (Decision & Instruction Generation Module), and Module 11 (System Integration & Mobile Processing Architecture)
Yumna Hussain	231149	Module 7 (Warning Sign Understanding), Module 8 (User Interaction & Voice Execution), Module 9 (Manual Interaction), Module 10 (Macro Navigation)

12. WBS and Gantt Chart

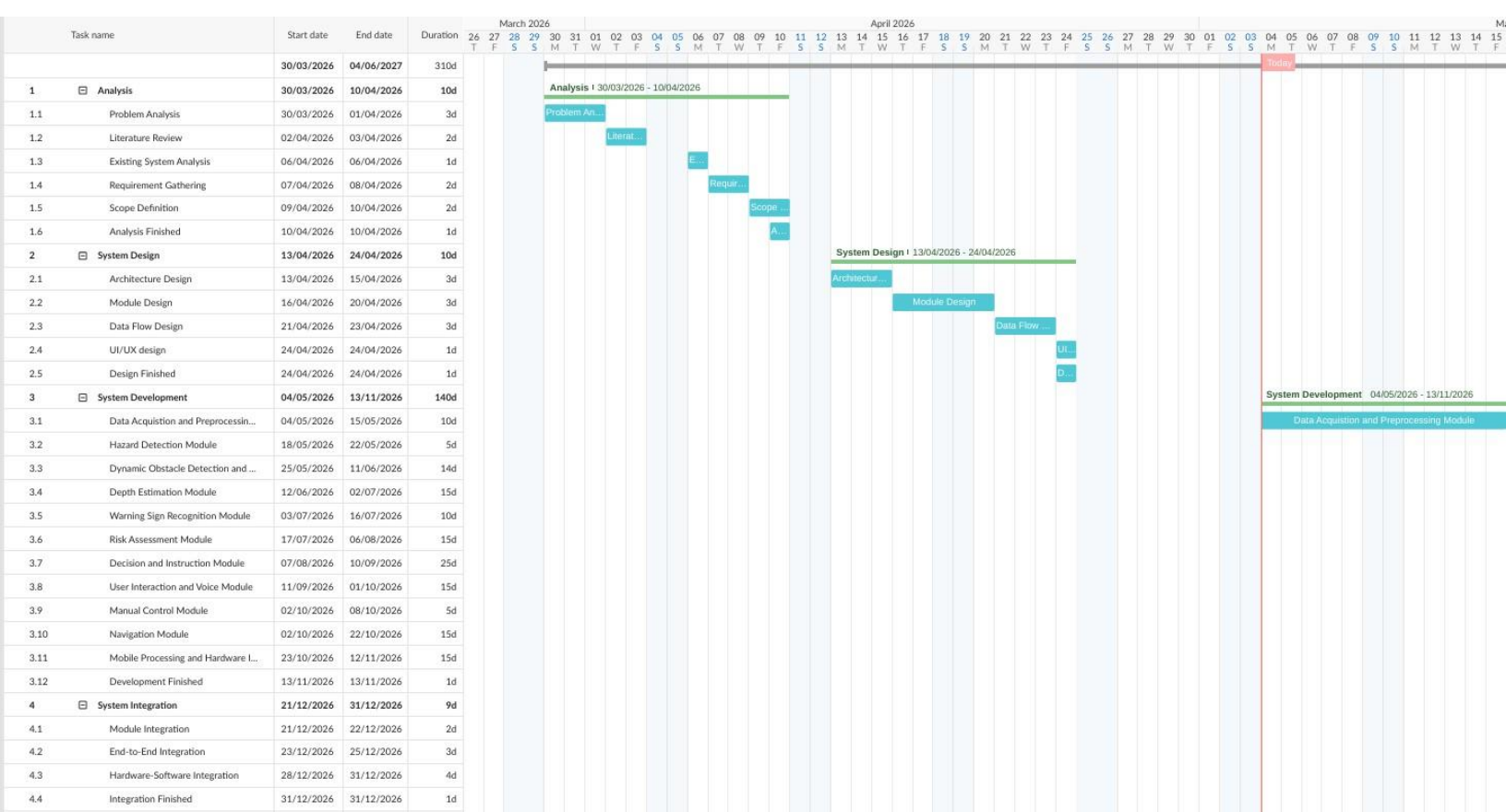
Task ID	Task Name		Duration	Resources
1	Analysis		2 weeks	Yousbah; Seerat; Yumna
2		Problem Analysis	3 days	Seerat
3		Literature Review	5 days	Yousbah
4		Existing System Analysis	3 days	Yousbah; Yumna
5		Requirement Gathering	4 days	Yumna
6		Scope Definition	2 days	Seerat
7		Analysis Finished	3 days	
8	System Design		2 weeks	Yousbah; Seerat; Yumna
9		Architecture Design	4 days	Yumna
10		Module Design	3 days	Yousbah; Seerat; Yumna
11		Data Flow Design	2 days	System Architect
12		UI/UX Design	4 days	UI Designer
13		Design Finished		

14	System Development		7 months 3 weeks	Yousbah; Seerat; Yumna
15		Data Acquisition & Preprocessing Module	2 week	Seerat
16		Hazard Detection Module	3 week	Seerat
17		Obstacle Detection & Tracking Module	3 weeks	Yousbah
18		Depth & Direction Module	3 weeks	Seerat
19		Warning Sign Recognition Module	2 week	Yumna
20		Risk Assessment Module	3 weeks	Yousbah
21		Decision & Instruction Module	5 weeks	Yousbah
22		User Interaction & Voice Module	3 week	Yumna
23		Manual Control Module	1 day	Yumna
24		Navigation Module	3 weeks	Yumna
25		Mobile Processing & Hardware Integration	3 weeks	Yousbah
26		Design Finished		

27	System Integration		2 weeks	Yousbah; Seerat; Yumna
28		Module Integration	4 days	Yousbah
29		End-to-End Integration	4 days	Yumna
30		Hardware-Software Integration	4 days	Seerat
31		Integration Finished		
32	Testing & Validation		1 month	Yousbah; Seerat; Yumna
33		Unit Testing	3 days	Seerat
34		Integration Testing	3 days	Yousbah; Seerat
35		System Testing	4 days	Yumna
36		Performance Testing	3 days	Yumna
37		Testing Finished		
38	Optimization		7 weeks	Yousbah; Seerat; Yumna
39		Performance Optimization	2 week	Yousbah ; Yumna
40		Accuracy Optimization	2 week	Seerat
41		Resource Optimization	1 week	Seerat
42		Latency Reduction	2 weeks	Yumna
43		Optimization Finished		

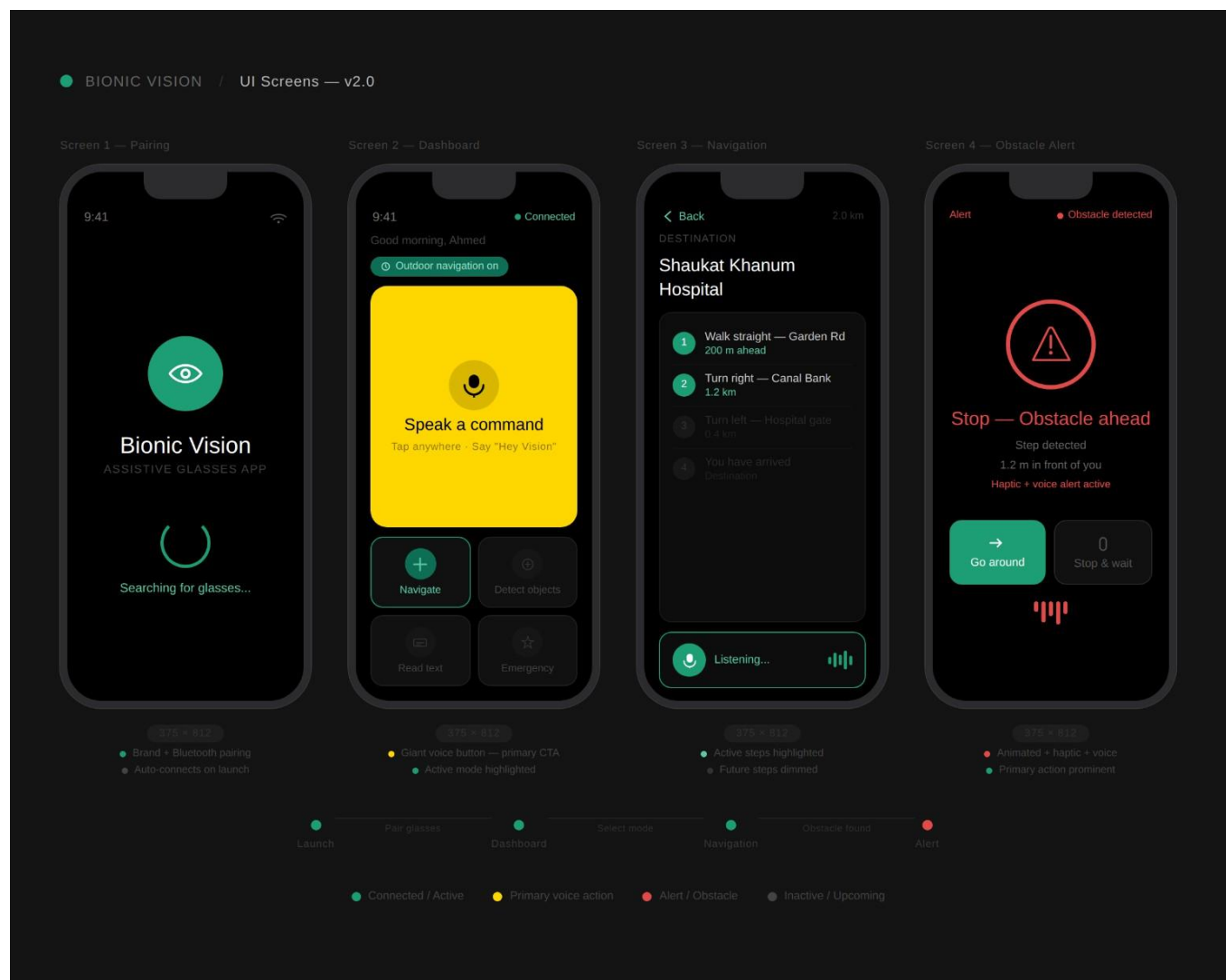
Proposal Document for Bionic Vision System

44	Documentation & Final Delivery		1 month	Yousbah; Seerat; Yumna
45		Inception Form	1 day	Yousbah; Seerat; Yumna
46		Scope Documentation	1 week	Yousbah; Seerat; Yumna
47		Final Report	1 week	Yousbah; Seerat; Yumna
48		Presentation Preparation	2 weeks	Yousbah; Seerat; Yumna
49		Final Demonstration	1 day	Yousbah; Seerat; Yumna



LINK:<https://app.ganttpro.com/shared/gantt/8d5b2456fbb0b965b9803e47d94a799ae05b8e52ed40f49b2501b1ca8e8241ed/2864270> (for complete GANTT CHART, use this link)

13. Mockups



Description: This mockup displays user interface of application. It is kept simple because a blind person doesn't have to interact with it.

14. References

- 1) Joshi, M., Shukla, A., Srivastava, J., Rastogi, M., Mujumdar, S., & Tripathi, H. (2023, August). Drishti: Visual navigation assistant for visually impaired. In *Journal of Physics: Conference Series* (Vol. 2570, No. 1, p. 012032). IOP Publishing.
- 2) Kotyan, S., Venkanna, U., Kumar, N., & Sahu, P. K. (2018, October). Drishtikon: An advanced navigational aid system for visually impaired people. In *2018 Conference on Information and Communication Technology (CICT)* (pp. 1-6). IEEE.
- 3) Potdar, K., Pai, C. D., & Akolkar, S. (2018). A convolutional neural network based live object recognition system as blind aid. *arXiv preprint arXiv:1811.10399*.
- 4) Varshney, A. S., Chougale, M. E., Patel, C. V., & Chauhan, M. D. (2026). Evaluating usability of “the smart vision glasses” for individuals who are visually impaired and totally blind. *Saudi Journal of Ophthalmology*, 40(1), 42-47.
- 5) Mushtaq, M. A., Shafiq, S., & Aziz, A. (2026). IoT & Computer Vision Based Smart Stick for Visually Impaired Persons.
- 6) Gudauskis, M., Žvironas, A., & Plikynas, D. (2026). Indoor Navigation Systems for Visually Impaired Persons: A Comprehensive Review of Technologies and Approaches. *Mobility of Visually Impaired People: Fundamentals and ICT Assistive Technologies*, 495-525.
- 7) Patil, S., Pawar, A., Pawar, P., Kulkarni, A., Gaikwad, V., Vairagar, S., ... & Metri, P. (2026, January). Smart Text-to-Speech System for Visually Impaired Users: A Hybrid AI-Based Solution. In *International Conference on Sustainable Innovation with Artificial Intelligence and Machine Learning 2025 (ICSIAIML 2025)* (pp. 287-296). Atlantis Press.
- 8) Malaekah, E., Alfahad, O., Bakouri, M., Gadallah, A., Selman, S., Al Rashdi, A., & Saied, H. (2026). Sound-based navigation system for visually impaired individuals. *Journal of Radiation Research and Applied Sciences*, 19(1), 102160.
- 9) Iqbal, K., Ali, S. S., Sajid, Z., Samad, M., Mubarak, L., & Ali, S. (2025). Advancement in Smart Vision Systems: A Computer Vision-Based Assistive System for Visually Impaired Individuals. *VAWKUM Transactions on Computer Sciences*, 13(1), 244-257.
- 10) Asch, A., & Mudrick, N. R. (2008). Visual impairment and blindness. In *Encyclopedia of social work*. Oxford University Press.

15. Plagiarism Report

Attach the Plagiarism report of your project scope document from library staff of turnitin tool <http://turnitin.co>